Product Stock Data Flow Diagram

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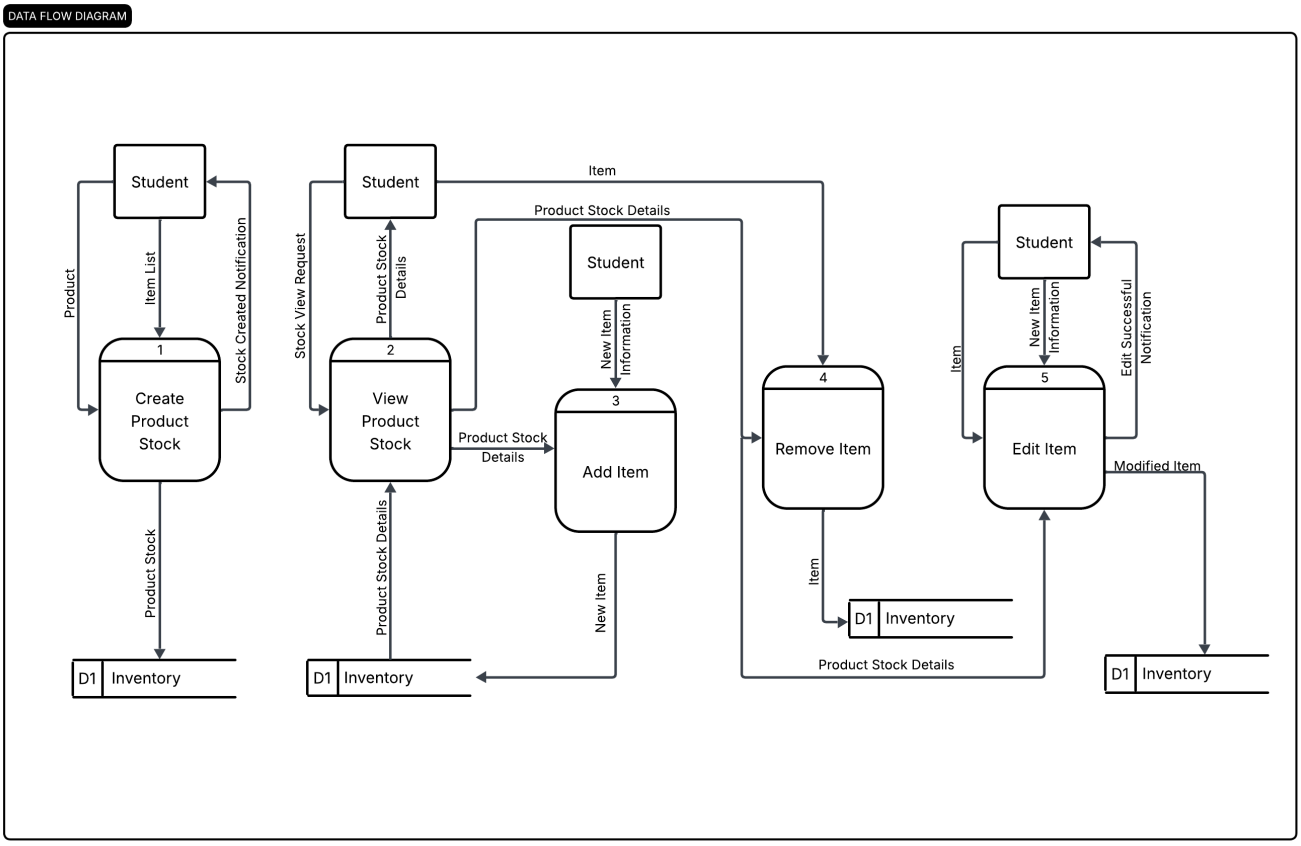
1. Purpose

Present the domain-oriented Data Flow Diagram (DFD) for the Product Stock feature of the Cafeteria Ordering System. The diagram represents how product stock data flows between external actors, processes, and data stores, supporting a shared team understanding, validation of requirements, and consistent implementation of the feature.

2. Diagram



3. External Entities

The following external entities interact with the Product Stock feature:

- Student – The primary actor of the feature. The student creates product stocks, views stock details, and manages individual stock items by adding, removing, or editing them.

4. Processes

The following processes represent the core business operations of the Product Stock feature:

4.1. Process 1: Create Product Stock

The student provides a product selection and an item list. In return, the student receives a stock creation confirmation once the stock has been registered.

4.2. Process 2: View Product Stock

The student provides a stock view request indicating the product they want to inspect. In return, the student receives the product stock details.

4.3. Process 3: Add Item

The student provides new item information for the product stock they are currently viewing. The

new item is then recorded in the Inventory.

4.4. Process 4: Remove Item

The student indicates the item they want removed from the product stock they are currently viewing. The item is then deleted from the Inventory.

4.5. Process 5: Edit Item

The student provides the item they want to edit along with the updated item information. In return, the student receives confirmation that the item was successfully modified.

5. Data Stores

The following data store supports the Product Stock feature:

- D1 Inventory – Stores all product stock records and individual item entries. It is read and written by all five processes within this feature.

6. Data Flows

The following table summarizes the data flows present in the diagram:

Flow Label	From	To	Description
Product Selection	Student	Create Product Stock	The product the student wants to create a stock for.
Item List	Student	Create Product Stock	The list of items the student wants to compose the stock with.
Product Stock	Create Product Stock	Inventory	The new stock record the student's submission produces in the inventory.
Stock Creation Confirmed	Create Product Stock	Student	The confirmation the student receives after the stock is successfully created.
Stock View Request	Student	View Product Stock	The product the student indicates they want to view the stock of.
Product Stock Details	Inventory	View Product Stock	The stock details retrieved for the product the student requested.

Flow Label	From	To	Description
Product Stock Details	View Product Stock	Student	The stock details delivered back to the student.
Product Stock Details	View Product Stock	Add Item	The product stock context the student is currently viewing, carried into the add operation.
New Item Information	Student	Add Item	The item information the student provides for the new item to be added.
New Item	Add Item	View Product Stock	The new item entry produced from the student's submission.
Item	Student	Remove Item	The item the student indicates they want removed.
Product Stock Details	Inventory	Remove Item	The stock context of the product the student is currently viewing.
Item	Student	Edit Item	The item the student indicates they want to edit.
New Item Information	Student	Edit Item	The updated item information the student provides.
Modified Item	Edit Item	Inventory	The updated item record produced from the student's submission.
Edit Successful Notification	Edit Item	Student	The confirmation the student receives after the item is successfully edited.

7. Identified Risks and Considerations

The following risks were identified during the development of this diagram:

- Ambiguity in stock requirements may result in missing or incorrect data flows during implementation.
- The Add Item, Remove Item, and Edit Item processes each reference the same Inventory data store. These are represented separately in the diagram for readability but refer to a single shared store.
- Edge cases such as empty stock, failed retrievals, or partial updates are not represented in this domain-level diagram and should be addressed at the implementation level.

- The diagram intentionally excludes UI and technical implementation details in order to maintain a domain-oriented perspective.